GAURI SANTHOSH NAIR

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SUMMARY

Third-year BSc Data Science & Mathematics student (GPA 3.82/4.00) at CHRIST (Deemed to be University), Bengaluru, researching **market risk and execution**. First author of a **Scopus-indexed paper (WSEAS Transactions on Systems and Control)** whose Hawkes jump-diffusion model beat a Poisson baseline in 253 of 259 high-frequency windows, and author of a no-look-ahead **VaR study for ERCOT supplier collateral** that cut the evening-ramp breach rate from 2.31% to 1.20%. Seeking a **Quantitative Research / Trade Strategy internship**.

EDUCATION

CHRIST (Deemed to be University), Bengaluru 2024 – 2028
BSc Data Science & Mathematics | GPA: 3.82 / 4.00
Relevant coursework: Stochastic Processes, Statistical Inference, Time Series Analysis, Numerical Optimisation, Operations Research, Financial Mathematics, Mathematical Modelling, Linear Algebra, Real Analysis, Algorithms, SQL

RESEARCH – CENTRE FOR MATHEMATICAL NEEDS, CHRIST (DEEMED TO BE UNIVERSITY)

Calibrating Tail Risk for ERCOT Supplier Collateral (Time-of-Day and Volatility-Filtered VaR) **2026**
Abstract accepted for presentation at ICETMOMTM 2026 (1st International Conference on Emerging Trends in Modelling and Optimization), Amity University, Noida; analysis updated since submission

- Sized a 99% VaR for a 50 MW supplier position and tested it walk-forward with no look-ahead on 31,119 five-minute real-time steps (HB_HOUSTON, 109 days) and 4,275 day-ahead steps (DA_NORTH, 2021–2025); scored with Kupiec and Christoffersen tests and pinball loss
- Showed that a static parametric VaR hides risk where it matters: a 0.83% overall breach rate masked **2.31% in the 17:00–22:00 ramp**, with clustered breaches (independence rejected, $p < 0.001$)
- EWMA-volatility-filtered historical simulation with hour-of-day quantiles cut the evening breach rate to **1.20%** (target 1%), halved breach severity (2.65× to 1.33× of VaR), lowered pinball loss from 18.46 to 7.38 and VaR as a share of notional from 49.7% to 29.3%; stable across nine decay and bucket settings
- Reported negative results as findings: no point-forecast edge over persistence, no day-ahead method beat full notional at 99%, no Hawkes clustering beyond time of day on 155 real days. Traced an earlier 2.6% forecasting edge to a unit-scaling bug, withdrew it and reframed the study as risk calibration

Endogenous Order Flow Dynamics and Jump Risk in Smart Order Routing: A Hawkes Jump-Diffusion Cross-Asset Study of Crypto Microstructure **2026**
Accepted, WSEAS Transactions on Systems and Control (pending final revision) · Co-author: Jinny Ann John

- First author. Embedded a bivariate Hawkes jump-diffusion process into the Cont-Kukanov smart order routing framework; on **259 high-frequency windows (26 days)** of real Binance tick data across BTC, ETH, SOL, BNB and XRP it beat a Poisson baseline in **253 of 259 (97.7%; ΔAIC up to 2,607)**
- Identified three stable execution regimes and an η-threshold effect for ETH; execution-cost savings up to **16.85% (SOL)**, with cost CV ≤ 0.001 in 99% of windows

Routing Failure in Nodal Energy Markets: A Hawkes Jump-Diffusion Framework with Congestion Penalty for Smart Order Routing in ERCOT **2026**
Manuscript · First author · Co-authors: Jossy P George, Vijayalakshmi A

- On 3.26M real 5-minute ERCOT settlement-point records (10 days), the correlation between transient impact and congestion rose from −0.014 at 08:00 to **0.97–0.99 at 19:00–20:00** in HB_NORTH and HB_WEST, consistent with the proposed congestion mechanism; Hawkes branching ratio on LMP spikes was 0.003 (near-zero self-excitation)
- In a matched, paired 300-episode backtest on congestion calibrated to real data, a multi-agent RL router cut simulated delivery cost vs TWAP by **2.5–6.7%**, but a lowest-congestion rule (9.4%) and an analytical allocation (7.8%) did better, and no strategy changed cost dispersion significantly (Brown-Forsythe $p > 0.5$)
- On real prices, choosing the previous hour's lowest-congestion hub beat an equal split by **$1.03/MWh** (day-block 95% CI $0.46–1.71)

EXPERIENCE

Zenith Metaplast Pvt. Ltd., Mysore | Business Intelligence Intern Apr 2026 – May 2026 (6 weeks)

- Designed a centralized Power BI reporting framework with shared data models and automated refresh pipelines, standardizing financial and operational dashboards across departments

Excelsoft Technologies Pvt. Ltd., Mysore | Corporate Finance Intern Apr 2023 – May 2023

- Built PyTorch forecasting models for quarterly revenue projections, cutting forecasting error by 20%; automated financial data processing and report generation in Python

PROJECTS & ACHIEVEMENTS

- **After-Hours Price Oracle (2026):** built a Python (NumPy) FastAPI service that estimates after-hours fair values for traditional assets from weekend crypto ticks, returning a Merton jump-diffusion confidence interval and an Almgren-Chriss-based risk haircut
- Selected to present the Sustainable Forex Trading Analytics Platform at the Industry-Academia Conclave to 300+ professionals across finance, banking, research and academia (2026)

SKILLS

Quantitative: VaR and backtesting (Kupiec, Christoffersen, pinball loss), Hawkes processes, jump-diffusion, EWMA volatility, MLE, Monte Carlo, walk-forward validation, time-series forecasting
Technical: Python (pandas, NumPy, PyTorch, scikit-learn), SQL, FastAPI, Excel (financial modelling), Power BI, Tableau, Git/GitHub, Cloud computing

U.S. Citizen (no visa sponsorship required) | Available January – July 2027